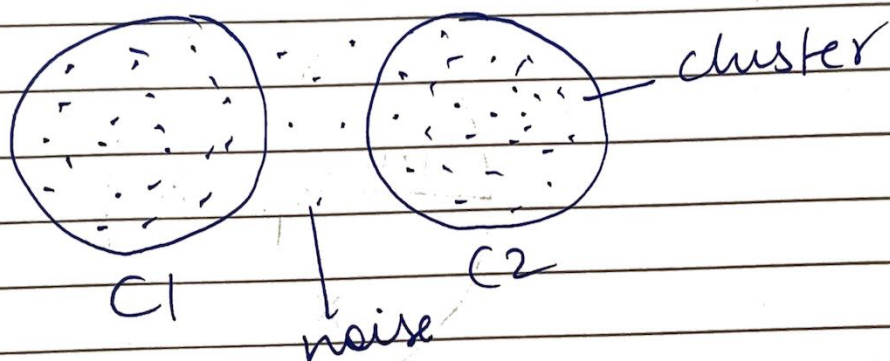


ch3

Density based clustering (DBSCAN) Technique

Video 1 : Density based clustering

Given a Dataset $D = \{x_i\}$ and you want to divide it into dense regions called clusters and sparse regions which are noise



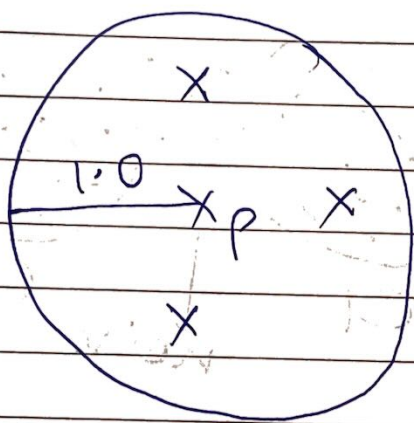
How to measure density?

- Min points
- Epsilon
- Core
- border
- noise

v2 Measuring density: Min points, Epsilon

These 2 are hyperparameters of DBSCAN.

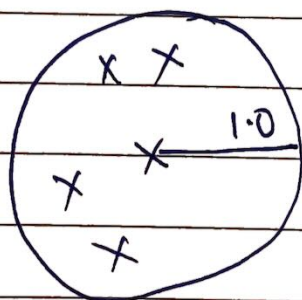
- 1) Density at a point P : # of points within a hypersphere of radius Epsilon around P .



density = 4 (inc. point P)

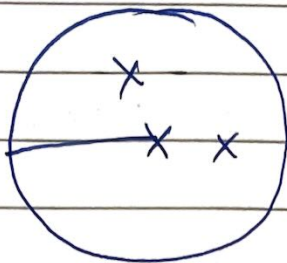
- 2) dense region: a hypersphere/circle of radius Eps that contains at least Minpts. points.

Let Minpts = 4



5 points
Hence dense region.

If 2 points in a regions.

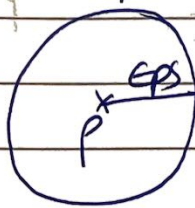


$3 < \text{Minpts}$

Hence sparse.

V3 Core, Border and Noise points

1) Core - If point p has $\geq \text{Minpts}$ in an Eps radius around it:



Core pt $\in$ dense region

2) Border pt(p) - ① If p is not a core pt, ie, p has $< \text{Minpts}$ points in Eps radius.

② $p \in \text{Neighbourhood}(Q)$

Q : core pts.

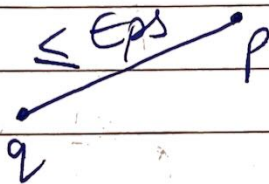
$$\text{dist}(p, Q) \leq \text{Eps}$$

3) Noise point: Anypt which is neither core point nor Border point is a noise pt.

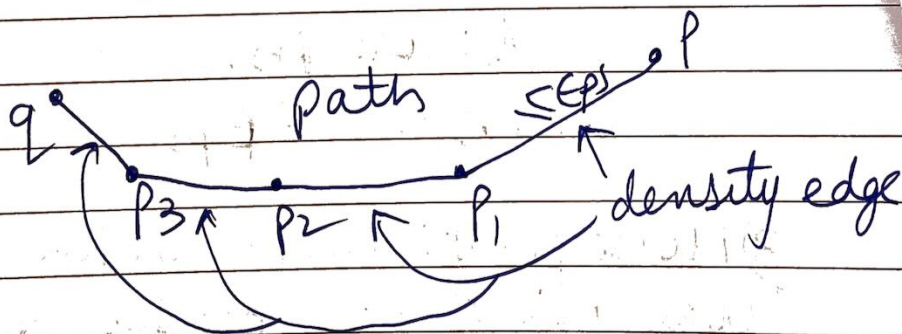
V4: Density edge and Density connected points

1) Density edge: connection.

If p and q are core points.
 If distance b/w p & $q \leq \epsilon_{ps}$
 then connect p and q call it
 as density edge.



2) Density connected pts: If p and q are core points, if there exists other points, such that there is density edge b/w each.
~~density core point~~



V5: DBSCAN Algorithm

Steps

① $\forall x_i \in D$

② Label core pt, or border pt or noise point.

To do this, do rangequery(x_i , D, ϵ)

③ Remove all noise points from your data. $\because$ they belong to sparse regions

loop ④ For each core point 'p' not assigned to a cluster

(i) Create a new cluster with p

(ii) Add all the points that are density connected to p into this new cluster

⑤ Each border point assign it to the nearest core point's cluster.

V6: Hyperparameters: MinPts and EpsA

1) MinPts

① Rule of thumb: $\text{MinPts} \geq d+1$

↓
dimension
of the
dataset

Typically: $\text{MinPts} \approx 2 * d$

② ~~Min~~ Larger MinPts if dataset is more noisy

③ MinPts parameter is chosen by domain expert.

If p has 10 pts within Eps radius then p is a noise point.

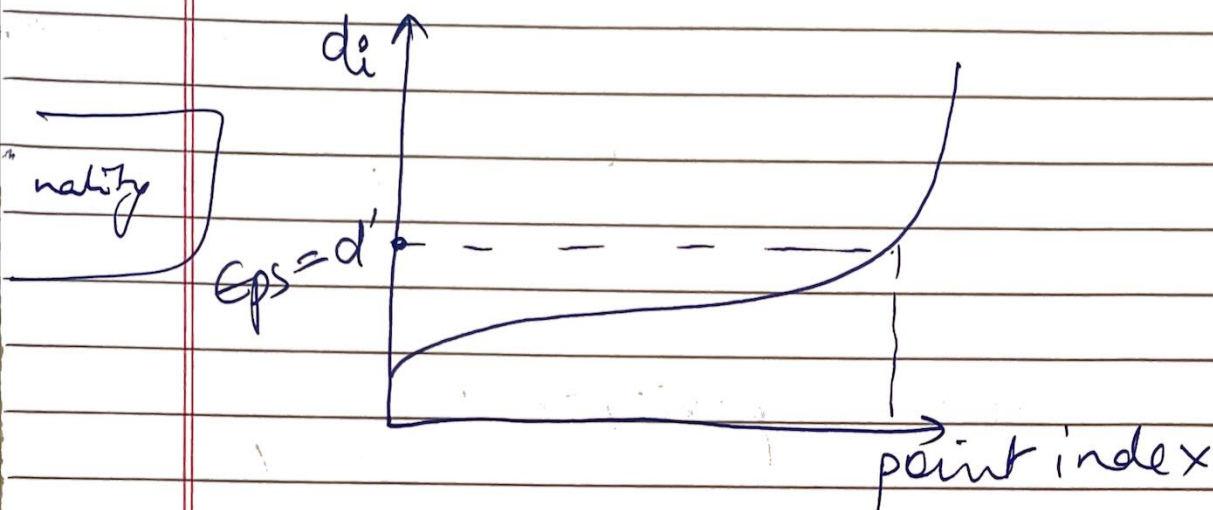
2) Eps

Let $\text{MinPts} = 4$, how to determine Eps

① for x_i compute d_i .

d_i is distance from x_i to the 4th NN of x_i

(b) Sort d_i 's in increasing order
elbow/knee method.



If high d_i then prob. that x_i is noisy is also high.

V7: Advantages and Limitations of DBSCAN

Adv.

- 1) Resistant to Noise
- 2) Can handle clusters

Disadv.

- 1) Varying densities of datapoints ~~give~~ does not give correct cluster.
- 2) ~~It~~ Very sensitive to hyperparameter.
- 3) With high dimensional data, there is curse of dimensionality.

V8: Time and Space complexity

$$\text{Time Complexity} = O(n \lg n)$$

n pts.

range or region
↓ query
{kd tree}

$$\text{Space Complexity} \rightarrow O(n)$$

as $d \ll n$

Works well when data is stored in database.

DBSCAN is not sensitive to NOISE.